Auditing Water-Balance Closure in a Soil–Plant Model with Interpretable Machine Learning

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Abstract

Scientific machine learning promises equations discovered from data, yet few discovery pipelines are first shown to recover a known answer, and fewer ask whether the quantities they are given close in the first place. A discovered relation is only as credible as the controls it has passed. A discovery protocol is developed in which a blind linear-invariant search must return the water balance before anything else is believed, with a permutation null and resampling to calibrate its uncertainty, and in which every partition used as a target is audited for closure. The protocol is applied to a large designed ensemble of simulations from a widely used soil–plant model, across experiments and soils that are withheld in turn. The search recovers the water balance with quantified uncertainty, and the closure audit then shows what the reported components leave out: an unattributed evaporative flux that is a substantial share of evapotranspiration, and two experiments in which stored water does not close. Because this flux changes the partition fractions that are used to compare management treatments, its identification matters to users of the model and of emulators built on it. A compact law with three terms predicts evapotranspiration in withheld experiments and withheld soils as accurately as gradient boosting, whereas drainage resists both, which locates the information boundary of annual flux data. Relations that look universal, such as irrigation compensation, prove regime-specific. The contribution is not a new law of nature but a template that tells a modeller which outputs can be trusted, which need an audit, and what a discovered equation must survive before it is called one. The results concern a process model and identify the questions that measured lysimeter and eddy-covariance data can settle.

Keywords. equation discovery; water-balance closure; interpretable machine learning; evapotranspiration partitioning; process-model auditing

List of symbols

Table 1: List of symbols, abbreviations and units used in the text.

Symbol	Meaning	Unit
$P_r,\ I,\ W$	rainfall, irrigation, water input $P_r + I$	mm yr^{-1}
$ET,\ T,\ E_s,\ E_c,\ R$	evapotranspiration and components; unattributed term, Eq. (2)	mm yr^{-1}
$D,\ \Delta S,\ Q_r$	drainage at 1 m, storage change, runoff	mm yr^{-1}
ε	closure residual, Eq. (3)	–
$\mathbf{F},\ \sigma_k,\ \mathbf{v}_k$	flux matrix, singular values and right singular vectors	–
$\mathbf{c}$	invariant coefficients, Eq. (8)	–
$n,\ \alpha$	van Genuchten shape and scale	–, cm^{-1}
TAW	total available water	mm
R^2_{cv}	cross-validated coefficient of determination, Eq. (6)	–
κ	irrigation compensation, Eq. (7)	–
E1–E17	numbered experiments of the Daisy ensemble	–
LOEO, LOSO	leave-one-experiment-out and leave-one-soil-out validation	–
SVD, PCA	singular-value decomposition; principal component analysis	–
GBM	gradient-boosting machine used as the flexible benchmark	–
DPF, NLF, IE	drainage fraction D/W; nitrogen-leaching fraction; irrigation efficiency	–

1. Introduction

Conservation of mass is the most reliable organising principle in hydrology, and it is the foundation of every water-balance model from the bucket to the Earth system model. Yet in practice the budget of a managed soil profile rarely closes exactly in the output of a process model: the fluxes that are reported separately (transpiration, soil evaporation, canopy interception) need not add to the evapotranspiration that enters the balance; change in storage is reported or implied but its size relative to the fluxes is seldom audited; and boundary fluxes at the base of a one-dimensional column are, by construction, definitions rather than measurements (Hillel, 1998; Refsgaard and Henriksen, 2004). At the same time, a strand of scientific machine learning has argued that conservation laws, governing equations and low-dimensional invariants can be *discovered* from data, using sparse regression over libraries of candidate terms (Brunton et al., 2016; Champion et al., 2019; de Silva et al., 2020), symbolic regression (Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020), neural networks that learn invariants (Iten et al., 2020; Liu and Tegmark, 2021), and physics-informed learning (Karniadakis et al., 2021; Raissi et al., 2019). In hydrology, deep learning has delivered impressive predictive skill (Kratzert et al., 2018, 2019; Shen, 2018), with an ongoing debate about what such models reveal about processes (Nearing

et al., 2021; Reichstein et al., 2019) and about whether interpretable models should be preferred for consequential decisions (Rudin, 2019). Enforcing conservation analytically in learned emulators is also an active topic (Beucler et al., 2021).

Three gaps limit confidence in discovery claims in soil-water science. First, discovery methods are rarely validated against a case with a known answer before being applied to an unknown. Second, a discovered relation is typically validated within the data from which it was found, without leave-one-*experiment*-out tests that ask whether the relation survives a change of regime, which is the relevant test of a "law" (Roberts et al., 2017). Third, negative results (where sparse laws fail to beat black-box benchmarks, or where a presumed constant is not constant) are rarely documented, so the reader cannot estimate the base rate of success (Kirchner, 2006; Oreskes et al., 1994).

A large designed ensemble of process-model runs, in which the true balance is known to hold up to the model's own storage term, is taken as a laboratory for method evaluation. A four-part pipeline is built (Fig. 1): (1) a linear-invariant search by singular-value decomposition of the flux matrix with a permutation null, used first as a positive control; (2) a unattributed-term search by auditing partitions that should sum to a total; (3) a manifold analysis of water-fate composition by compositional PCA (Aitchison, 1986; Egozcue et al., 2003); and (4) interpretable sparse regression over a dimensionless library with LOSO and LOEO validation against a gradient-boosting benchmark (Friedman, 2001; Hastie et al., 2009; Tibshirani, 1996). One fixed benchmark, cross-validation grouping and stopping rule are used for all targets, and every law that failed is reported.

The paper asks what a blind method can and cannot recover from a flux matrix. A linear-invariant search must first return the water balance with calibrated uncertainty, which sets the credibility of everything that follows. Accounting terms that the model's own output columns leave open are then sought, and the number of independent directions governing the partition of water among fates is counted. Compact, dimensionally sensible laws are tested against a flexible machine-learning model in *unseen experiments and unseen soils*. The last aim is to separate relations that hold across regimes from those that only appear universal.

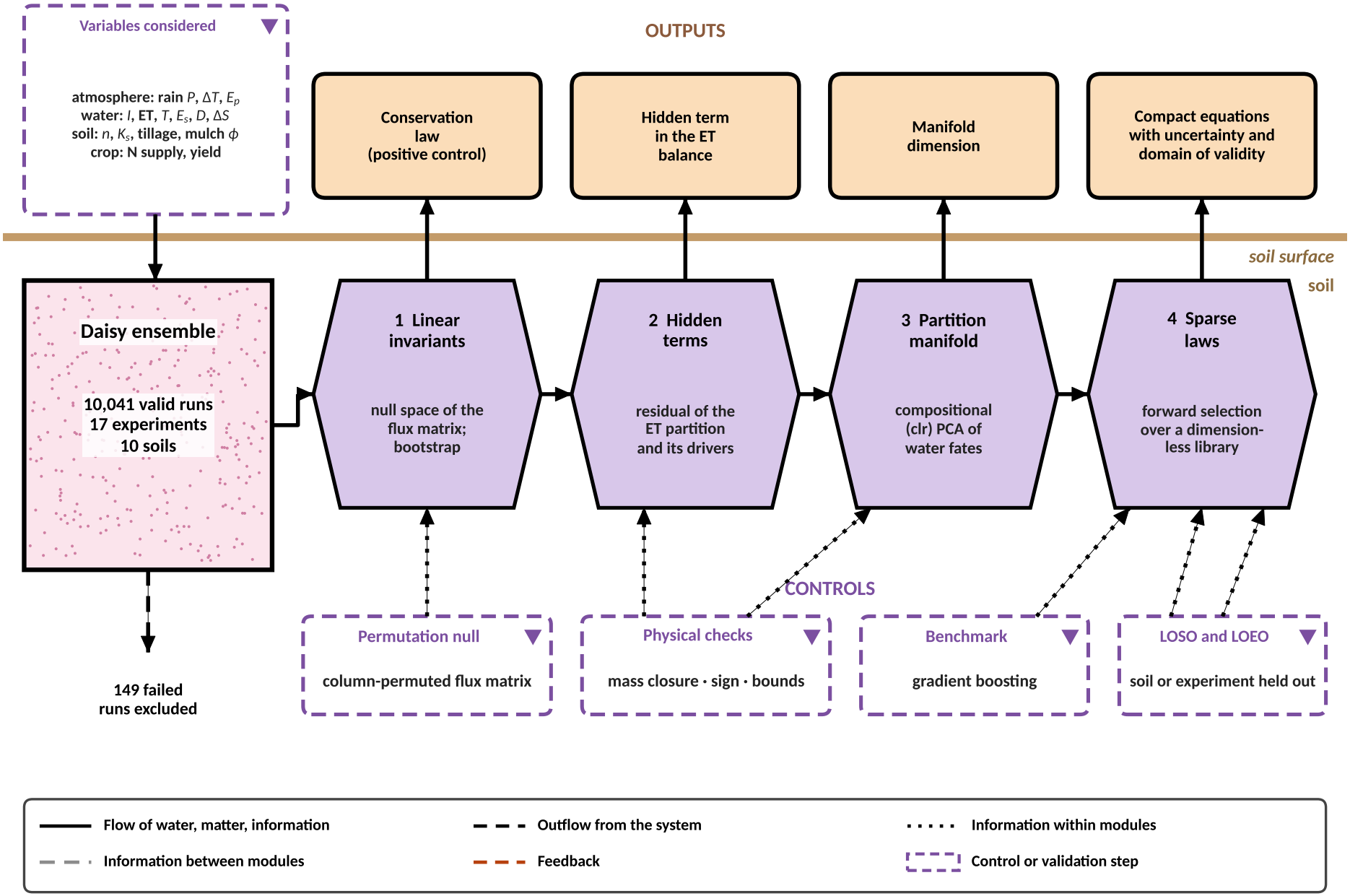


Figure 1: Pipeline for interpretable discovery. The pipeline recovers the known water balance as a positive control, audits partitions, measures the dimension of the partition manifold and fits sparse laws with LOSO/LOEO validation against gradient boosting. Controls (dashed purple boxes) are shown at the bottom and outputs at the top; purple hexagons are processing blocks, the pink stippled box is the data store, peach boxes are outputs and dotted lines are flows of information. The 149 runs excluded as numerical failures are analysed in Section 2.2.

Study design and levels of evidence. The unit of observation is the annual-mean water balance of one simulated run, and the runs of an experiment are correlated because they share a design; for this reason validation withholds whole experiments and whole soils. Four levels of evidence are kept apart. Computational reproducibility, the regeneration of every reported number from archived code and data, is established. Predictive validity is assessed by leave-one-experiment-out and leave-one-soil-out scoring against a flexible benchmark, with a nested evaluation of the selection step. Mechanistic validity is limited to the process model, so that the unattributed flux is a property of the model's accounting. Field transferability is not tested, and the conclusions are stated as questions that measured data can settle.

2. Materials and methods

2.1 Data

The Daisy (Hansen et al., 2012; Holbak et al., 2026) virtual-experiment ensemble is used, as described in the companion paper: 10 190 runs in 17 experiments (E0–E16), ten soils (S01–S10, van Genuchten $n = 1.23$–2.68, Table 3), 20-year Taastrup weather; 149 runs failed and are excluded (Section 2.2), and 10 041 valid runs are analysed. Each run contributes 20-year mean annual rainfall P_r, irrigation I, evapotranspiration ET, transpiration T, soil evaporation E_s, canopy evaporation E_c, drainage at 1 m D, and (derived) crop yield, nitrogen leaching and irrigation efficiency. Experiments differ in structure: single-soil (coarse loam) lattices (E8, E11, E12, E14, E15), all-soil factorials and lattices (E9, E16) and one-factor series (E1–E4). The distributions of the key variables are in Table 2.

Table 2: Descriptive statistics of the 10 041 valid runs (mm yr^{-1}; yield in Mg ha^{-1} not shown).

Variable	Mean	SD	25 %	Median	75 %
Rain P_r	417	143	297	464	464
Irrigation I	180	191	0	178	268
Water input $W = P_r + I$	597	251	464	654	722
Evapotranspiration ET	546	196	386	611	671
Drainage D (1 m)	60	106	3	34	78
Unattributed R (Eq. (2))	164	96	103	129	223

Table 3: Soil parameters used as predictors (effective van Genuchten values from the project documentation; *TAW* in mm to 1 m).

ID	Texture	α (cm^{-1})	n	*TAW*
S01	Coarse loam	0.0961	1.353	130
S02	Coarse sand	0.0536	1.600	105
S03	Sand	0.1450	2.680	60
S04	Loamy sand	0.1240	2.280	75
S05	Sandy loam	0.0750	1.890	95
S06	Loam	0.0360	1.560	150
S07	Silt loam	0.0200	1.410	220
S08	Sandy clay loam	0.0590	1.480	160
S09	Clay loam	0.0190	1.310	175
S10	Silty clay loam	0.0100	1.230	195

2.2 Failed runs and why they are excluded

Of the 10 190 runs, 149 (1.5 %) are flagged as failed in the ensemble and are excluded from every analysis. Because exclusion that is not random can bias a discovery study, the pattern of failure was examined before the runs were dropped, and the argument is set out in four parts.

Failures are concentrated in three soils and in specific experiments. Of the 149 failures, 129 (87 %) occur in the coarse sand S02, 13 in the sand S03 and 7 in the silty clay loam S10; the other seven soils have none (Fig. 2a). In S02, 24.9 % of the 519 runs fail. In S03 all 13 failures belong to the initial-moisture experiment E9, and in S10 all 7 belong to the nitrogen–water and dose experiments (E13, E1). The three soils lie at the extremes or in the anomalous part of the van Genuchten range ($n = 1.60$ for S02, $n = 2.68$ for S03, the steepest retention curve, and $n = 1.23$ for S10, the flattest), which is where the unsaturated conductivity function changes most rapidly with suction.

Within a failing soil, failure follows the size of the water pulse. In S02, the fraction of failed runs rises with the irrigation depth applied per event in experiment E10: none of the 20 runs at 1 and 2 mm fails, 10 % at 3 mm, 50 % at 8–10 mm and all of the 30 runs at 15, 20 and 30 mm (Fig. 2b). It also depends on the rain scale: 69 % of the S02 runs at a scale of 1.5 fail (22 of 32) and 12 % at 0.5 (5 of 42), against none of the 90 runs at intermediate scales 0.6–1.4 (Fig. 2c). Runs that share the same weather, the same soil and the same code, and differ only in how much water arrives at once, therefore fail or succeed in a systematic order. This is the behaviour of a state-dependent numerical difficulty and not of a corrupted input, which would hit runs irrespective of their forcing.

The outputs of failed runs are physically impossible. In valid runs the annual drainage at 1 m is never below -17.5 mm (median in valid S02 runs, 134 mm). In failed runs it is negative in 65 % of cases and falls to -882 mm (Fig. 2d), and the ratio of evapotranspiration to water input ($P + I$) ranges from 0.54 to 4.34, with a median of 1.09, so that in some runs more water leaves the profile than entered it. A net upward flux of hundreds of millimetres through the lower boundary of a free-draining column, or evapotranspiration several times the water input, cannot be produced by the physics of the model; it is the signature of a solver that has not converged, in which a mass error is absorbed in the boundary or storage terms.

The alternatives are less consistent with the pattern. A fault in the weather or in the run definition would not depend on soil and on dose in this ordered way; a failure from over-wetting in the field, such as ponding or runoff, is a physical process that the model is built to represent and would not give negative drainage; and a random job failure would spread evenly over soils. Numerical non-convergence of the Richards-equation solver in a stiff regime (a very permeable, quickly draining column that receives a large input, or a column with an extreme retention curve) accounts for the concentration in soils, the order with water input and the impossible outputs together.

Treatment of the excluded runs. The failed runs are interpreted as numerical failures of the soil-water solver under stiff conditions, not as information about soil hydrology, and they are excluded for that reason; the interpretation follows from the pattern of failure described above. Two consequences follow. The exclusion removes a quarter of the runs of S02, preferentially in the wettest and driest forcing and at the largest doses, so results for this soil, and any law fitted to it, apply to the part of the factor space in which the solver succeeded. The other nine soils are essentially unaffected (at most 2.5 % of runs in S03), and the conclusions that rest on across-soil comparisons were tested for sensitivity to the exclusion by re-analysis. With S02 removed (9 651 runs) the three-term ET law is re-estimated with coefficients 0.652 for $\ln W$, -0.297 for n and 0.029 for $\ln(1 + I)$, against 0.646, -0.299 and 0.029 for the full ensemble, with LOEO $R^2 = 0.897$ and LOSO $R^2 = 0.904$ (full ensemble 0.898 and 0.904), and the unattributed term R is 30 % of ET (163 mm yr^{-1}; full ensemble 30 % and 164 mm yr^{-1}). With S02 and S03 both removed (9 145 runs) the scores are 0.893 and 0.898 and R is 30 % (165 mm yr^{-1}). The exclusion therefore does not drive the across-soil results.

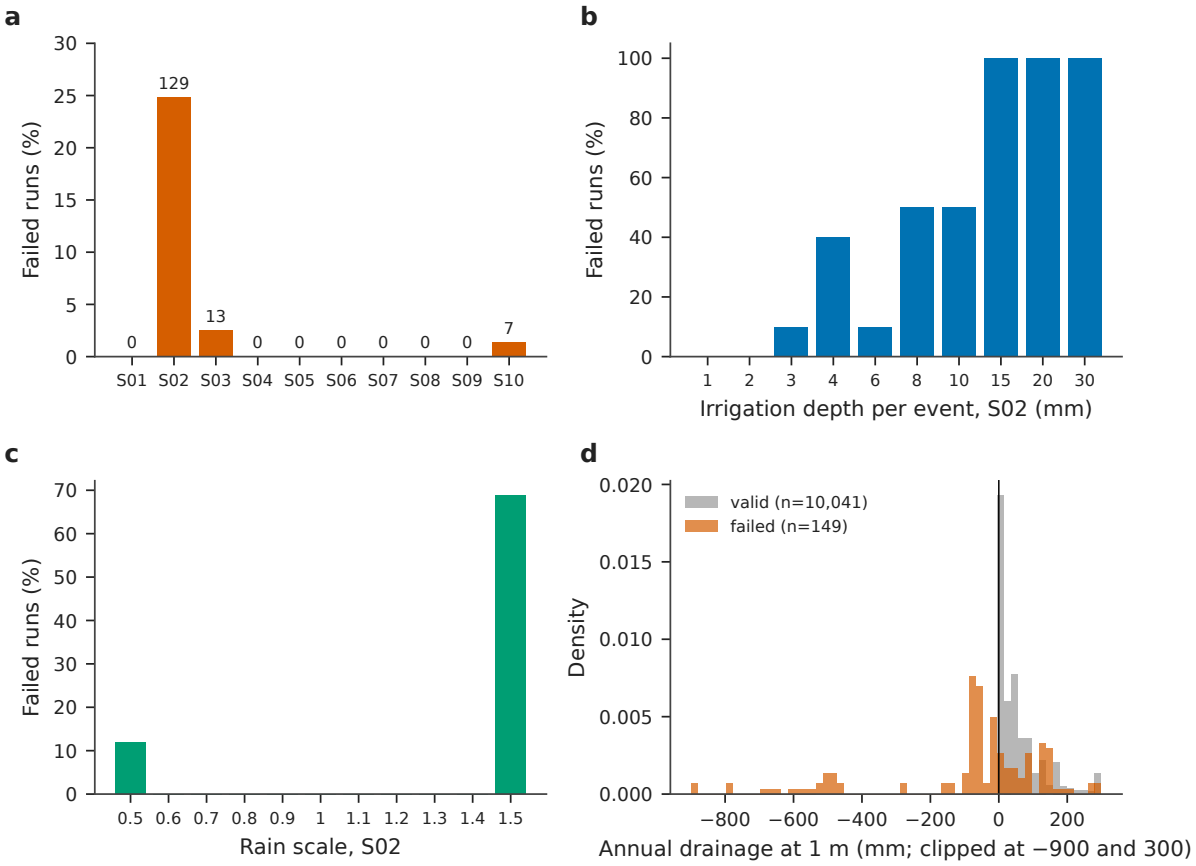


Figure 2: Pattern of the 149 failed runs. (a) Failed runs per soil, as a percentage of the 519 runs per soil (counts above bars). (b) Failed fraction in S02 against irrigation depth per event in E10. (c) Failed fraction in S02 against rain scale. (d) Annual drainage at 1 m for valid and failed runs, normalised as densities; negative drainage of several hundred millimetres occurs only in failed runs.

2.3 Governing balance and the flux matrix

For each run the mean annual water balance is

$$P_r + I \;=\; ET + D + \Delta S + Q_r, \qquad ET = T + E_s + E_c + R, \tag{1}$$

where ΔS is storage change, Q_r runoff, and R is defined as the remainder of ET after the three reported components,

$$R \;=\; ET - T - E_s - E_c. \tag{2}$$

Equation (2) defines R without ascribing a process. Let $\mathbf{F} \in \mathbb{R}^{N\times q}$ be the matrix whose rows are runs and whose columns are the fluxes (rain, irrigation, ET, T, E_s, E_c, D). If a linear invariant $\sum_j c_j F_{ij} = 0$ holds for all runs, then $\mathbf{c}$ lies in the null space of $\mathbf{F}$. Define $W = P_r + I$ and the

closure residual

$$\varepsilon \;=\; \frac{W - ET - D}{W} \;\approx\; \frac{\Delta S + Q_r}{W}. \tag{3}$$

2.4 Linear-invariant search by SVD with a permutation null

The singular-value decomposition $\mathbf{F} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ with singular values $\sigma_1 \geq \cdots \geq \sigma_q$ identifies near-null directions as the right singular vectors $\mathbf{v}_k$ with small σ_k. Each column is first divided by its root-mean-square value so that no flux dominates through its units; $\mathbf{v}_k$ is mapped back to mm units and normalised so that the ET coefficient equals -1. Because singular values depend on the scale of the columns, significance is judged against a *column-permutation null*: each column of $\mathbf{F}$ is permuted independently across runs (destroying cross-column dependence but preserving each marginal), and the smallest singular value under the null is recorded; an invariant is accepted only if $\sigma_{\min}$ lies far below the null value. Uncertainty of the coefficients is estimated by a nonparametric bootstrap over runs (resampling 10 041 rows; 95 % percentile intervals). The test is repeated (a) on the four-flux subsystem (P_r, I, ET, D), which is the set that appears in the balance; (b) on the seven-column matrix; and (c) excluding E8 and E9, whose initial states are deliberately away from equilibrium. It is emphasised that this is a *positive control*: the expected answer is $\mathbf{c} = (1, 1, -1, -1)$ up to scale and the pipeline is judged by how closely it recovers it, and by whether it fails in a diagnosable way.

Any decomposition that should sum to a total is audited by computing the residual. The audit covers (a) Eq. (2), reporting R, its share R/ET by experiment, and its dependence on drivers by regression of $\ln R$ on candidate drivers (irrigation, soil, mulch capacity, temperature, evaporation factor); and (b) the storage closure ε of Eq. (3) by experiment. The audit is descriptive: it identifies where the budget does not close, and tests candidate attributions by whether R responds to the drivers that would be expected under each.

The partition of water input into fates (transpiration, soil evaporation, canopy evaporation, unattributed R, drainage) is compositional: components are positive and sum to a constant, so Euclidean statistics are inappropriate (Aitchison, 1986; Egozcue et al., 2003). Writing the composition as $\mathbf{x} = (x_1, \ldots, x_5)$ with $x_i = \text{flux}_i / \sum_j \text{flux}_j$, the centred log-ratio (clr) transform is applied

$$\operatorname{clr}(\mathbf{x})_i = \ln x_i - \frac{1}{5}\sum_{j=1}^{5} \ln x_j, \tag{4}$$

and PCA (Wold et al., 1987) to the transformed data. Drainage is zero or negative in 14 % of

runs, so zeros are replaced by a floor δ (in mm yr^{-1}) before the closure to unit sum; because the results depend strongly on δ, results are reported for $\delta \in \{10^{-3}, 0.1, 1, 5, 10\}$. Because log-transformations are inherently sensitive to near-zero values, the sensitivity to the floor is mapped across four orders of magnitude and not fixed at a single value, so that the stability of the drainage axis can be judged independently of the floor selected. The number of components needed to explain the variance is a data-driven estimate of the dimension of the partition manifold.

For a target y (log of a flux), a library $\boldsymbol{\Theta}$ is built of dimensionless or log-transformed candidates (Table 4). The model is

$$\ln y \;=\; \beta_0 + \sum_{k \in \mathcal{S}} \beta_k \, \theta_k + \epsilon, \qquad \mathcal{S} \subseteq \{1, \ldots, p\}, \tag{5}$$

with the active set $\mathcal{S}$ chosen by forward selection: at each iteration the candidate that maximises the leave-one-experiment-out cross-validated R^2 is added, and the search stops when the improvement is below 0.003. The cross-validated R^2 is

$$R^2_{\mathrm{cv}} = 1 - \frac{\sum_i (\ln y_i - \widehat{\ln y}_i^{(-g(i))})^2}{\sum_i (\ln y_i - \overline{\ln y})^2}, \tag{6}$$

where $\widehat{\ln y}_i^{(-g(i))}$ is the prediction for run i from a model fitted with all runs of its group $g(i)$ held out. Groups are experiments (LOEO, 17 folds) or soils (LOSO, 10 folds). LOEO is the demanding test: it asks whether a law fitted on 16 experimental designs predicts the 17th, which differs in which factors vary and in what range. The root-mean-square error in mm and the mean absolute percentage error after back-transformation are also reported. The benchmark is a histogram-based gradient-boosting regressor (Friedman, 2001; Pedregosa et al., 2011) with 250 iterations on the whole library, validated under the same groupings. Caveat: because forward selection uses the LOEO score, the reported LOEO R^2 of the selected model is mildly optimistic (selection bias); LOSO values, which were not used for selection, provide a partly independent check, and coefficient stability across folds is reported.

Table 4: Candidate library $\boldsymbol{\Theta}$ (log-transformed or dimensionless).

Symbol	Definition	Role
$\ln P_r$	log rainfall	climate input
$\ln(1+I)$	log irrigation	management input
$\ln W$	log water input $P_r + I$	total supply
$\ln E_p$	log soil-evaporation factor	evaporative demand
ΔT	temperature shift (°C)	warming
n	van Genuchten shape parameter	pore-size distribution
$\ln TAW$	log total available water	storage capacity
$\ln \alpha$	log van Genuchten scaling	air-entry scale
mulch	VFF×CAP (mulch lattice)	residue layer
I/W	irrigation fraction	managed share
$\ln(W/TAW)$	dimensionless supply-to-storage	wetness index

Suppose the true model is $\mathbf{Fc} = \boldsymbol{\eta}$ with $\boldsymbol{\eta}$ a small run-wise residual (here $-\Delta S - Q_r$ for $\mathbf{c} = (1, 1, -1, -1)$). The SVD gives the best rank-$(q-1)$ approximation of $\mathbf{F}$ (Eckart–Young), and the right singular vector $\mathbf{v}_q$ minimises $\|\mathbf{Fv}\|_2$ over unit vectors; the corresponding singular value is $\sigma_q = \|\mathbf{Fv}_q\|_2$. Because the columns are all measured with error (here, the model's own discretisation and storage noise), $\mathbf{v}_q$ is a total-least-squares solution, in which errors in all variables are treated symmetrically; this is preferable to regressing one flux on the others, whose coefficients would be attenuated and depend on the choice of dependent variable. When η is uncorrelated with the columns, $\sigma_q^2 \approx \|\boldsymbol{\eta}\|^2$, so $\sigma_q/\sqrt{N}$ is the root-mean-square residual per run in the units of the scaled columns. Before the decomposition each column is divided by its root-mean-square value (so that no flux dominates by its units), and the singular vector is mapped back to mm units afterwards; the singular values are therefore dimensionless, and for the four-flux subsystem $\sigma_q/\sqrt{N} = 5.9/\sqrt{10041} = 0.059$, i.e. a residual of about 6 % of a typical column scale. A caveat is that the matrix is not mean-centred: centring would remove the constant direction, but a conservation law $\sum c_j F_j = 0$ has no intercept, so the uncentred matrix is the correct object.

Permuting each column independently preserves marginal distributions and destroys all linear relations between columns, so the smallest singular value under permutation, σ_q^{perm}, measures how small σ_q would be if the columns were unrelated but had the same scales. A ratio $\sigma_q/\sigma_q^{\text{perm}} = 5.9/33.3 = 0.18$ for the four-flux matrix shows that the null direction reflects genuine dependence. Because fluxes are positively skewed and share a common experiment structure (for example rainfall is identical across many runs), the permutation null is conservative in one respect (it breaks the experimental clustering) and liberal in another (it does not respect the clustering when permuting). Runs are therefore also bootstrapped and intervals reported.

1. Initialise $\mathcal{S} = \emptyset$, best score $s^* = -\infty$.
2. For each candidate $k \notin \mathcal{S}$, fit Eq. (5) on the training groups and compute R^2_{cv} (Eq. (6)) for $\mathcal{S} \cup \{k\}$ with 17 LOEO folds.
3. Let k^* be the maximiser; if $R^2_{\text{cv}}(\mathcal{S} \cup \{k^*\}) > s^* + 0.003$, add k^* and update s^*; otherwise stop.
4. Refit on all data to report coefficients; evaluate LOSO using the same terms.

Table 11 shows the path for the ET law: $\ln W$ alone gives $R^2_{\text{cv}} = 0.801$, adding n raises it to 0.895 and adding $\ln(1 + I)$ to 0.898. The third term improves the score by only 0.003, at the stopping threshold, so a two-term law $\ln ET = f(\ln W, n)$ is almost as good and is the safer summary.

The compensation coefficient κ measures the share of a rainfall change that is offset by irrigation. For rainfall change ΔP_r with other factors fixed,

$$\kappa \;=\; -\frac{\Delta I}{\Delta P_r}. \tag{7}$$

In single-soil lattices $\mathrm{d}W/\mathrm{d}P_r = 1 - \kappa$ is estimated by ordinary least squares; in E16, paired differences are formed of rain scale 0.5 and 1.5 across the 32 combinations of other factors within each soil, and bootstrap the pooled estimate (500 resamples).

All analyses are scripted in Python (Harris et al., 2020; Hunter, 2007; Pedregosa et al., 2011; Virtanen et al., 2020); scripts and intermediate results are in the project repository (Data, Code) and mapped in Appendix B. Model evaluation follows the recommendations of Moriasi et al. (2007) and Bennett et al. (2013); R^2 is reported on the log scale unless stated. No test data were used for model selection other than through the declared cross-validation.

3. Results

3.1 The pipeline recovers the water balance (positive control)

The singular spectrum of the seven-flux matrix falls steeply, with a smallest singular value of 5.5 compared with about 33 under column permutation (Fig. 3a), and the corresponding right singular vector places weight on rain, irrigation, ET and drainage with opposite signs for inputs and outputs (Fig. 3b). For the four-flux subsystem the null direction, scaled to $c_{ET} = -1$, is

$$\mathbf{c} = (c_{P_r}, c_I, c_{ET}, c_D) = (1.037,\ 0.957,\ -1,\ -1.036), \tag{8}$$

with bootstrap 95 % intervals 1.031–1.045 for rain, 0.940–0.974 for irrigation and -1.052 to -1.023 for drainage, and a smallest singular value of 5.9 against 33.3 under permutation. The pipeline therefore recovers the conservation law $P_r + I - ET - D \approx 0$ with every coefficient within 5 % of the exact value, and the permutation null shows that the null direction is not an artefact of scale. *However, the bootstrap intervals do not contain the exact values*: rain $\pm$ 1 lies below the rain interval, and irrigation lies above its interval. The deviation reflects the nonzero storage term and other unreported fluxes: excluding E8 and E9, whose initial moisture is deliberately unbalanced, gives $(1.021,\ 0.969,\ -1,\ -1.019)$ with a smallest singular value of 3.0, closer to the exact balance but still not exact.

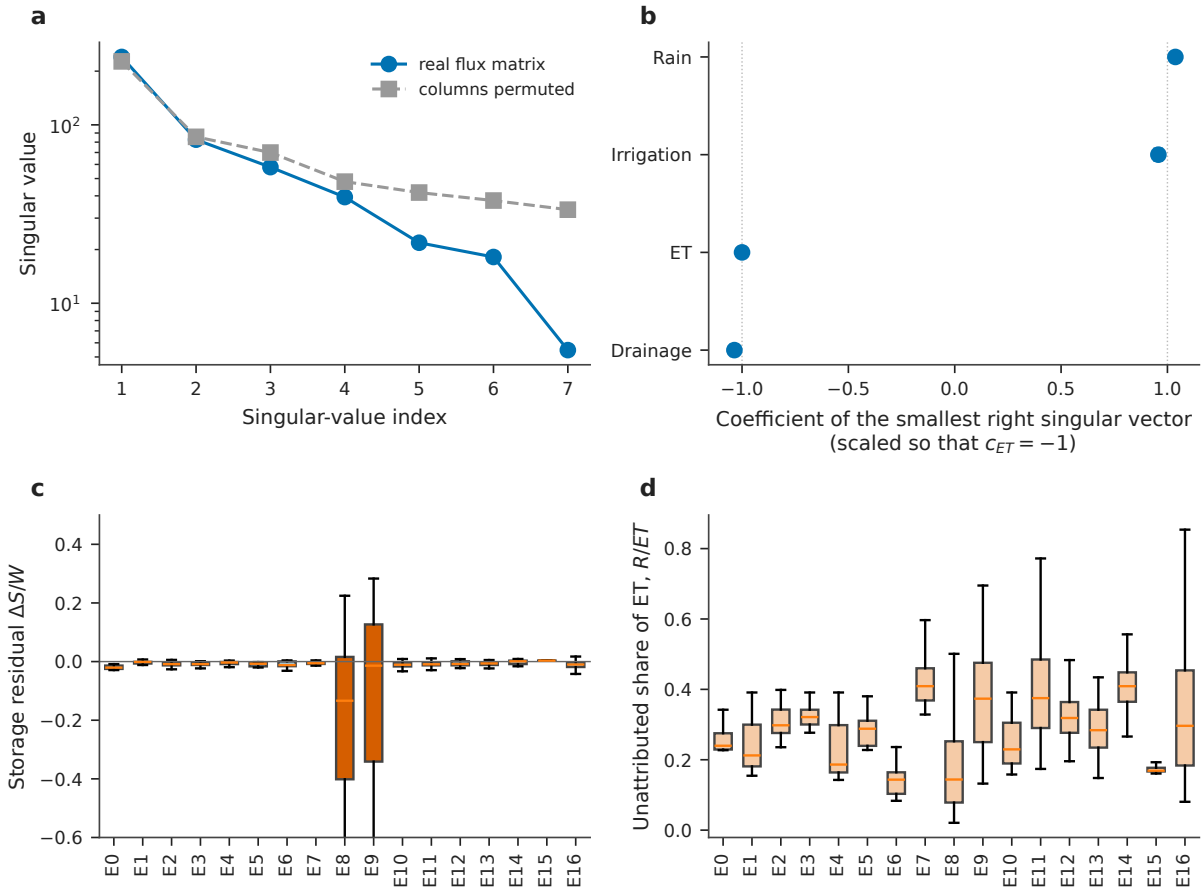


Figure 3: Linear invariants. (a) Singular values of the seven-flux matrix for the real data and for column-permuted data. (b) Coefficients of the smallest right singular vector (scaled so $c_{ET} = -1$) with bootstrap intervals. (c) Storage residual $\Delta S/W$ by experiment: E8 and E9 do not close. (d) Unattributed share of ET, R/ET, by experiment.

The result illustrates both the promise and the discipline needed for law discovery. The method finds the law that is there, without being told which variables are inputs and which are outputs; but its coefficients should not be over-interpreted, because the third decimal of a coefficient reflects what the data do *not* close, not what the law is.

Fitting the four-flux null vector separately within each experiment (Table 5) tests whether the pooled vector Eq. (8) hides heterogeneity. In 11 of the 14 experiments with irrigation all coefficients are within about 5 % of the exact values $(1, 1, -1, -1)$ (E11 is at 5.1 % for drainage; E16, with the extreme corners of the factorial, gives 0.901 for irrigation and -0.919 for drainage, E6 gives 0.961, 0.927 and -0.869; and E12 gives drainage -1.116 are the exceptions). The residual scale $\sigma_q/\sqrt{N}$ is 0.001–0.017 in the well-closing experiments and 0.081 in E16, 0.030 in E6, 0.166 in E8 and 0.219 in E9. E0 (no irrigation variation within the 10 baseline runs), E8 and E9 (rainfed) have a constant zero irrigation column and the invariant is formed from three fluxes; for E8 the drainage coefficient is $+15.5$, a sign of an ill-conditioned decomposition in an experiment in which storage dominates the balance, which is a cautionary example of how an unconstrained search can return a numerically valid but physically meaningless vector. The method therefore identifies its own failure by the magnitude of $\sigma_q/\sqrt{N}$ and by physical-plausibility checks on the coefficients, which is recommended as a rule.

Table 5: Four-flux null vector fitted within each experiment (scaled to $c_{ET} = -1$); "–" marks a flux with no variation in the experiment (irrigation in E0, E8, E9). $\hat{\sigma} = \sigma_q/\sqrt{N}$ is the residual scale in units of the column root-mean-square.

Experiment	Runs	c_{P_r}	c_I	c_D	$\hat{\sigma}$
E0	10	1.025	–	−1.041	0.004
E1	189	1.009	0.969	−0.979	0.017
E2	197	1.022	0.983	−1.027	0.004
E3	200	1.016	0.995	−1.012	0.004
E4	96	1.006	0.989	−0.983	0.003
E5	300	1.016	0.993	−1.027	0.002
E6	193	0.961	0.927	−0.869	0.030
E7	200	1.008	0.992	−0.990	0.003
E8	1000	1.241	–	15.496	0.166
E9	987	1.182	–	−1.249	0.219
E10	954	1.026	0.972	−1.038	0.010
E11	1000	0.998	1.024	−1.051	0.004
E12	1000	1.016	1.005	−1.116	0.005
E13	975	1.016	0.984	−1.008	0.004
E14	1000	0.982	1.035	−0.974	0.001
E15	1000	1.009	0.966	−0.971	0.003
E16	740	1.018	0.901	−0.919	0.081

The closure residual ε (Eq. (3)) has a median absolute value of 0.9 % of water input over all valid runs and 0.65 % excluding E8 and E9 (95th percentile 2.3 %), so for 15 of the 17 experiments the mass balance holds to within a few percent. In E8 and E9, which vary initial moisture, the median absolute residual is 16 % of W, with the residual reaching ± 0.3–0.6 in the tails (Fig. 3c).

This is expected, because these experiments deliberately impose a non-equilibrium initial storage that is released over the 20-year window, but it matters for discovery: a conservation law fitted without storage will appear to be violated in exactly the experiments where hydraulic memory is strongest. A global law should therefore include storage explicitly or be fitted only on near-equilibrium experiments, and the two cases give different coefficients (Eq. (8) and the text following it).

3.2 A unattributed term in the ET partition

The ET components that Daisy reports do not add up to ET. The remainder R (Eq. (2)) averages 164 mm yr^{-1}, with an interquartile range of 103–223 mm yr^{-1}, and equals 30 % of ET (median 29 %; range 2–88 %). Its share varies between experiments from 14 % (E6, E8) to 41 % (E7, E14; Fig. 3d, Table 6). The dependence of R on management is informative (Fig. 4). In E4 (methods), R is 105 mm yr^{-1} for every subsurface-drip configuration (5–60 cm), 183 for surface and 166 for overhead irrigation, so that *where* water is applied matters while the drip depth does not (Fig. 4a). In the rainfed E9 lattice, $R = 50.9 + 0.121\,P_r$ (Fig. 4b), i.e. about 12 % of rainfall, and in E11 (coarse loam)

$$\ln R = 5.75 - 0.355 \ln P + 0.863 \ln E_p + 0.136\,\Delta T, \qquad R^2 = 0.979, \tag{9}$$

where P is the rain scale, E_p the soil-evaporation factor and ΔT the temperature shift, so that R scales almost linearly with the soil-evaporation factor, decreases with rain scale (elasticity -0.36) and increases with warming. The mulch experiments E7 and E14 have the highest median shares (41 %) and the cross-experiment sparse model for $\ln R$ selects $\ln(1 + I)$, n, the mulch capacity and ΔT.

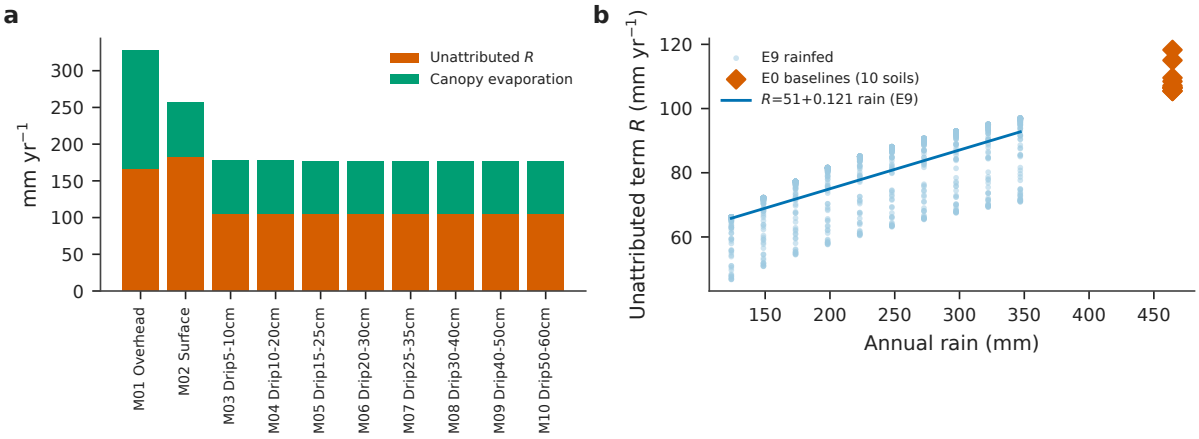


Figure 4: The unattributed term R. (a) R and canopy evaporation in E4 by irrigation method. (b) R against annual rain in E9 (rainfed, light points), with the linear fit and the ten E0 baselines.

Table 6: Unattributed term R by experiment. median R (mm yr^{-1}), median share of ET, and median storage residual $\Delta S/W$.

Experiment	Runs	R (mm)	R/ET	$\Delta S/W$
E0 Baselines	10	107	0.24	−0.021
E1 Doses	189	118	0.21	−0.001
E2 Trigger pressure	197	168	0.30	−0.009
E3 Sensor depth	200	202	0.32	−0.009
E4 Methods	96	103	0.19	−0.003
E5 SHP dynamics	300	162	0.29	−0.007
E6 Mediterranean	193	70	0.14	−0.012
E7 Mulching	200	226	0.41	−0.006
E8 Rain intensity×moisture	1000	39	0.14	−0.133
E9 Soil moisture×rain	987	81	0.37	−0.014
E10 Amount, timing, soil	954	118	0.23	−0.011
E11 Temperature×E_p×rain	1000	296	0.38	−0.009
E12 Sensor, trigger, rain	1000	218	0.32	−0.008
E13 N×water	975	125	0.28	−0.006
E14 Mulch×rain	1000	262	0.41	0.001
E15 Method, depth, amount	1000	103	0.17	0.003
E16 Edge cases	740	157	0.30	−0.010

What is R? The data do not identify it, but they constrain candidates. A process that is independent of drip depth, larger under surface and overhead application, proportional to the soil-evaporation factor, increased under mulch and weakly increasing with warming is consistent with evaporation from the *surface layer* (mulch or litter and ponded water) that is not reported in the soil-evaporation or canopy-evaporation columns. That reading is a hypothesis; the model code has not been inspected to confirm which Daisy state variables sum into the model's ET output, and a software-level audit of the output definitions would be required to turn the hypothesis into

a statement. The practical consequence is nonetheless clear: any analysis that treats $T + E_s + E_c$ as ET (as many reduced-form ET-partitioning analyses do) omits about 30 % of the flux, and the partition fractions reported for the ensemble depend on how R is allocated.

A sparse model for R across all experiments has limited out-of-sample skill. The forward-selected model $\ln R = 3.35 + 0.194\ln(1+I) + 0.492\,n + 0.145\,\mathrm{cap} + 0.238\,\Delta T$ has an in-sample $R^2 = 0.61$ on the log scale but a leave-one-experiment-out error of 0.51 in $\ln R$, equal to a level-scale R^2 of only 0.30 (RMSE 81 mm) versus 0.50 (RMSE 69 mm) for gradient boosting. Within a single experiment, Eq. (9) is excellent; across experiments it is not, which implies that R depends on interactions among irrigation method, mulch state and demand that the library does not contain.

3.3 A dominant drainage axis, with its variance share quantified by bootstrap

The log-ratio transform requires strictly positive components, and drainage is zero or negative (net capillary rise) in 1 429 of the 10 041 runs (14 %) and below 1 mm yr^{-1} in 1 947. The treatment of these zeros is therefore not a detail. The first analysis replaced all components below 10^{-3} mm yr^{-1} by that floor, which gave explained variances of 88.6 %, 7.3 % and 3.3 % for the first three components. Because $\ln(10^{-3})$ lies far from the logs of the other components, the runs with zero drainage then dominate the first axis. Repeating the analysis with floors of 0.1, 1, 5 and 10 mm yr^{-1} gives a first-component share of 74.3, 62.7, 56.5 and 52.3 % (Fig. 5a, Table 7). The result that a single drainage axis dominates is robust, since the loading of drainage on PC1 stays between $+0.82$ and $+0.88$ for every floor and also when only runs with drainage above 5 or 20 mm are used (PC1 share 57 % and 52 %), but its *magnitude* is a function of the data-handling choice and the figure of 88.6 % should not be quoted as a property of the system. The number of components needed to explain 90 % of the variance is two to three in all cases (three components carry 92–98 % for floors from 10^{-3} to 10 mm, with 97.3 % at the 1 mm floor adopted for the loadings, Fig. 5b,c).

With the 1 mm floor, the first component (62.7 %) is a drainage-versus-evaporation axis: drainage loads $+0.84$, soil evaporation -0.37, unattributed R -0.34, transpiration -0.20 and canopy evaporation $+0.06$. Its scores correlate with the drainage fraction D/W at Spearman $\rho = 0.98$, with $\ln(W/TAW)$ at 0.48 and with n at 0.45. The second component (24.8 %) separates canopy evaporation ($+0.77$) from soil evaporation (-0.56), and the third (9.8 %) opposes R ($+0.83$) to soil evaporation (-0.43) and transpiration (-0.26). The choice of the 1 mm floor

was made *after* seeing the sensitivity and is justified only as a value close to the resolution of annual flux accounting; the full sensitivity is reported so that readers can judge. The practical conclusion is that the drainage fraction is, to a first approximation, the leading summary of water partitioning (its score correlation is 0.98), and that a second axis, the canopy-versus-soil evaporation split, is needed for a complete description.

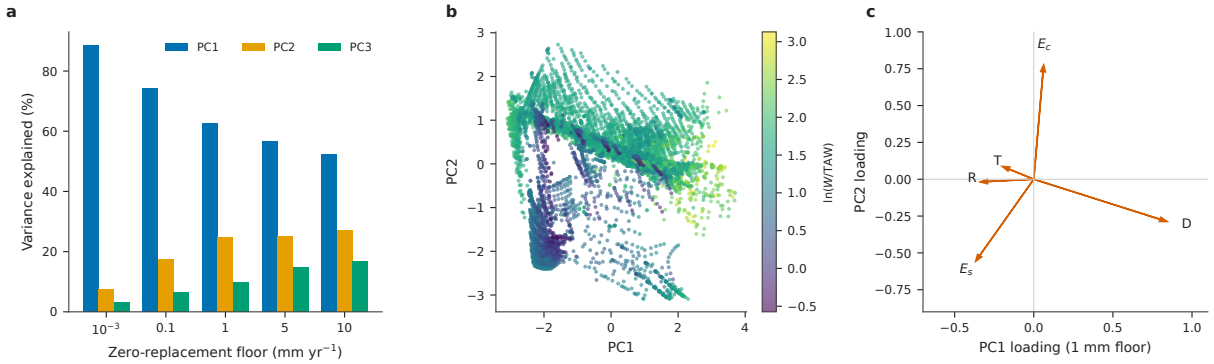


Figure 5: Partition manifold and its sensitivity to zero handling. (a) Variance explained by the first three clr-PCA components for five zero-replacement floors. (b) Scores on the first two components (1 mm floor) coloured by $\ln(W/TAW)$. (c) Loadings of the five fates on the first two components.

Table 7: Sensitivity of the clr-PCA to the zero-replacement floor (variance explained, %).

Floor (mm yr^{-1})	PC1	PC2	PC3	PC4
10^{-3}	88.6	7.3	3.3	0.8
0.1	74.3	17.3	6.6	1.7
1	62.7	24.8	9.8	2.6
5	56.5	25.1	14.8	3.6
10	52.3	27.1	16.8	3.8

Averaged over the 15 near-equilibrium experiments, the five fates (Fig. 6a) show the expected gradient: drainage takes 34.5 % of water input in sand (S03), 28.5 % in loamy sand and 5.4 % in coarse loam (S01), 8.5 % in silty clay loam, while transpiration rises from 19 % in sand to 31–33 % in the loams. The unattributed term, in contrast, is remarkably constant across soils: R takes 22–26 % of W in nine of ten soils (31 % in coarse loam, which has the mulch and temperature lattices) and its share of ET is 23–39 % (Fig. 6c), the largest in sand, where there is least else to evaporate. Since R is almost soil-independent in share of W, it behaves as a surface-driven term rather than a soil-hydraulic one, consistent with the surface-layer hypothesis. The stacked fractions sum to 0.96–1.01 of W (the remainder being storage change and runoff). Panel b is a reminder that coarse loam contributes 4 419 of the near-equilibrium runs while each other soil contributes 290–419.

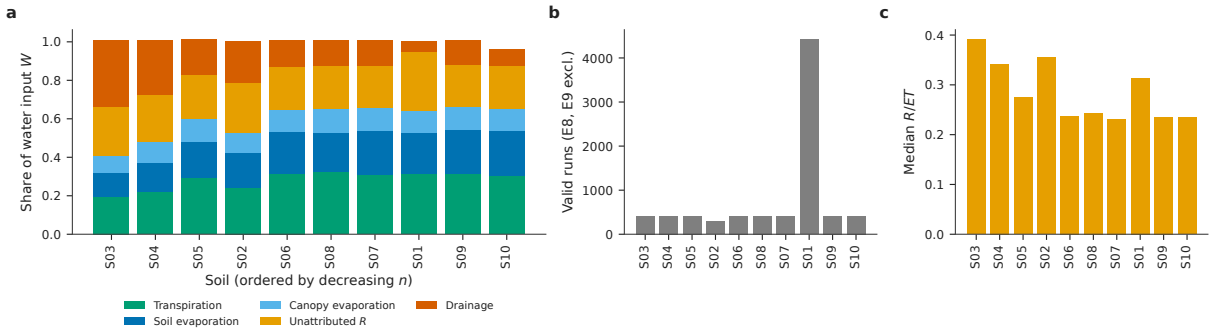


Figure 6: Water fates by soil in the 15 near-equilibrium experiments (E8, E9 excluded). (a) Mean fate shares of water input W, soils ordered by decreasing n. (b) Number of valid runs per soil. (c) Median share of the unattributed term in ET.

3.4 A compact evapotranspiration law with out-of-sample skill

Forward selection over the library selects three terms for ET (Table 13): $\ln W$, n and $\ln(1+I)$, giving

$$\ln ET = 2.503 + 0.646\ \ln W - 0.299\, n + 0.029\ \ln(1+I), \tag{10}$$

with ordinary-least-squares standard errors 0.027, 0.0051, 0.0045 and 0.0011 (which understate uncertainty because runs are not independent) and in-sample $R^2 = 0.908$. Leave-one-experiment-out $R^2 = 0.898$ and leave-one-soil-out $R^2 = 0.904$, against 0.876–0.879 (LOEO) and 0.881–0.894 (LOSO) for gradient boosting in two independent runs, so the three-term law matches or exceeds the benchmark under both groupings (Fig. 7a,b). The RMSE in leave-one-experiment-out prediction is 68.8 mm yr^{-1} (MAPE 9.2 %). The fitted coefficients are stable across the 17 folds (Fig. 7c; coefficient means 2.51, 0.645, -0.298, 0.029 with standard deviations of 0.077, 0.017, 0.017, 0.0023).

Equation (10) has a straightforward physical reading. The exponent on W (0.646) means that ET rises sublinearly with water supply, the signature of an energy- or storage-limited regime also encoded in Budyko-type curves (Budyko, 1974; Milly, 1994; Zhang et al., 2001): a 10 % increase in supply raises ET by 6.5 %, and the remainder goes to drainage, runoff and storage. The coefficient on n (-0.299) implies a factor $\exp(-0.299 \times 1.45) = 0.65$ in ET between the most uniform (S03, $n = 2.68$) and the most graded (S10, $n = 1.23$) soil at the same W: soils that drain freely retain less water for evaporation, in agreement with the hydraulic-memory results of the companion paper. The small positive coefficient on $\ln(1+I)$ (0.029) indicates that, at equal total input, a larger irrigated share raises ET slightly, plausibly through the wetting of the surface that increases R and E_s. The law is *not* a fundamental conservation law, nor a model of mechanism;

it is a compact summary of how the model's ET responds across 17 designs.

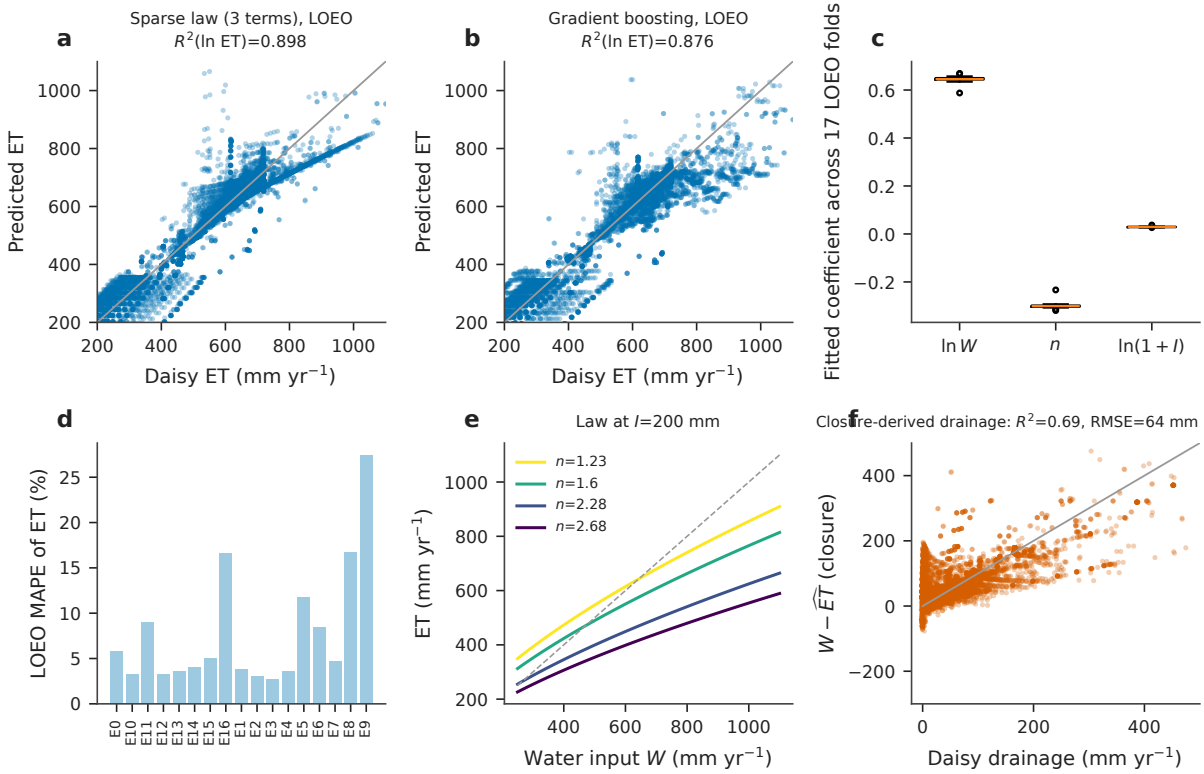


Figure 7: Compact ET law. (a) Leave-one-experiment-out predictions of the three-term law versus Daisy ET. (b) Gradient-boosting benchmark. (c) Fitted coefficients across the 17 LOEO folds. (d) LOEO MAPE by experiment. (e) The law at $I = 200$ mm for four soils (grey dashed: ET $= W$). (f) Drainage derived from closure, $W - \widehat{ET}$, versus Daisy drainage ($R^2 = 0.69$, RMSE 64 mm; E8 and E9 excluded).

The 17 experiments were resampled with replacement (2 000 cluster-bootstrap resamples) and the cross-validated predictions of the law were compared with those of a gradient-boosting model that uses the same three inputs ($\ln W$, n, $\ln(1 + I)$; LOEO $R^2 = 0.866$). The paired difference in R^2 is $+0.032$ with a 95 % interval of 0.004–0.087, so the law is better than a flexible learner given the same information in this resampling, with the caveat that the law's terms were selected with the same folds. With the *full* library the benchmark reaches 0.876–0.879, and the two are regarded as comparable rather than claiming superiority.

The error is not uniform (Table 8). Mean absolute percentage errors are 2.6–5.8 % in most one- and two-factor experiments, but 16.6 % in E16 (edge cases: extreme combinations), 16.7 % in E8 and 27.4 % in E9, the two experiments that violate storage closure, and 11.7 % in E5 (time-varying soil hydraulic properties, where a single static n is wrong by construction). The law is therefore valid where the water balance is near equilibrium and soil hydraulics are static; it should not be extrapolated to initial-state perturbations or to dynamic soil structure.

Table 8: Leave-one-experiment-out performance of Eq. (10) by experiment: MAPE (%) and mean error (predicted minus Daisy, mm yr^{-1}).

Experiment	Runs	MAPE (%)	Mean error (mm)
E0 Baselines	10	5.8	−25.2
E1 Doses	189	3.8	9.8
E2 Trigger pressure	197	3.0	−6.6
E3 Sensor depth	200	2.6	−8.0
E4 Methods	96	3.6	9.4
E5 SHP dynamics	300	11.7	−63.8
E6 Mediterranean	193	8.5	41.5
E7 Mulching	200	4.7	10.6
E8 Rain intensity	1000	16.7	−11.3
E9 Soil moisture	987	27.4	12.9
E10 Amount, timing	954	3.3	1.3
E11 Temperature, E_p, rain	1000	9.0	−62.6
E12 Sensor, trigger, rain	1000	3.3	−15.2
E13 N, water	975	3.5	−3.7
E14 Mulch, rain	1000	4.0	7.5
E15 Method, depth, amount	1000	5.0	29.0
E16 Edge cases	740	16.6	34.5

Table 9 gives the closure residual ε for every experiment. In 15 experiments the median absolute residual is below 2.1 % of W and the 95th percentile below 4 % (E6, the Mediterranean-pollution scenarios, is the exception at 10.7 %); in E8 and E9 the median absolute residual is about 16 % and the 95th percentile exceeds 100 % of W. For E8 the mean (signed) residual is negative, −13 %: the profile is gaining water, that is, the run begins drier than the equilibrium state and refills. The law Eq. (10) was fitted on water inputs from 124 to 2 762 mm yr^{-1} (ET of 95–1 600 mm yr^{-1}), n from 1.23 to 2.68, in a temperate climate, and it should be applied only within this domain and with near-equilibrium storage.

Table 9: Closure residual $\varepsilon = (W - ET - D)/W$ by experiment (per cent of water input): median signed, median absolute and 95th percentile of absolute residual.

Experiment	Runs	Median ε	Median $\lvert\varepsilon\rvert$	95th pct. $\lvert\varepsilon\rvert$
E0	10	-2.06	2.06	2.9
E1	189	-0.12	0.33	1.2
E2	197	-0.87	0.87	2.3
E3	200	-0.87	0.87	2.0
E4	96	-0.33	0.34	1.2
E5	300	-0.74	0.74	1.9
E6	193	-1.19	1.57	10.7
E7	200	-0.57	0.57	1.2
E8	1000	-13.35	15.76	105.8
E9	987	-1.40	16.87	108.5
E10	954	-1.08	1.17	2.5
E11	1000	-0.93	0.94	1.8
E12	1000	-0.82	0.83	1.8
E13	975	-0.64	0.64	2.3
E14	1000	0.09	0.40	1.5
E15	1000	0.34	0.34	0.4
E16	740	-1.01	1.22	3.8

The ensemble is heavily imbalanced: 5 519 of the 10 041 runs (55 %) are coarse loam (S01) because the large lattices E8, E11, E12, E14 and E15 use only that soil, so soil-level behaviour in the other nine soils rests on about 400–520 runs each. The LOEO residuals of Eq. (10) by soil (Table 10) show that the law over-predicts ET in the fine soils (mean error $+22$, $+20$ and $+31$ mm in S08, S09 and S10, where positive means over-prediction relative to Daisy, since errors are predicted minus Daisy) and has the largest percentage errors in sand (15.1 %) and loamy sand (13.7 %), the soils for which n is most extreme. A single shape parameter therefore does not remove all soil dependence, and the MAPE of about 7.5–15 % by soil is consistent with the overall 9.2 %.

Table 10: LOEO residuals of the ET law by soil (predicted minus Daisy, mm yr^{-1}). MAPE: mean absolute percentage error; RMSE in mm yr^{-1}.

Soil	Texture	Runs	Mean error	MAPE (%)	RMSE
S01	Coarse loam	5519	-10.7	7.5	58.9
S02	Coarse sand	390	15.8	8.6	69.8
S03	Sand	506	-13.6	15.1	63.8
S04	Loamy sand	519	-4.8	13.7	57.9
S05	Sandy loam	519	-3.8	10.9	56.1
S06	Loam	519	-0.2	9.9	77.5
S07	Silt loam	519	0.4	10.6	106.2
S08	Sandy clay loam	519	22.2	10.2	57.8
S09	Clay loam	519	19.7	9.9	92.8
S10	Silty clay loam	512	31.2	11.6	107.4

Table 11: Forward-selection path for the ET law. LOEO R^2_{cv} after each added term; and for other targets (first terms).

Target	Term added	R^2_{cv} (LOEO)
ln *ET*	1 ($\ln W$)	0.801
	2 (n)	0.895
	3 ($\ln(1+I)$)	0.898
$\ln T$	1 ($\ln W$)	0.581
	2 (n)	0.641
$\ln E_s$	1 (n)	0.070
	2 (mulch)	0.237
	3 ($\ln P_r$)	0.308
$\ln E_c$	1 ($\ln W$)	0.641
	2 ($\ln P_r$)	0.651
	3 (I/W)	0.668
ln yield	1 ($\ln W$)	0.531
	2 ($\ln(1+I)$)	0.549
	3 ($\ln P_r$)	0.599
	4 (mulch)	0.604
ln DPF	1 ($\ln(W/TAW)$)	0.193
	2 ($\ln \alpha$)	0.260
	3 (I/W)	0.422
	4 (ΔT)	0.433

If E8 and E9, which break closure, are removed from both fitting and validation (15 experiments, 8 054 runs), the same three terms give LOEO $R^2 = 0.882$, LOSO $R^2 = 0.881$, RMSE 65.9 mm and MAPE 5.8 %, so the law is more accurate (relative error 5.8 % versus 9.2 %) in the domain for which it is intended. The marginal contribution of the terms is clearer in this subset: $\ln W$ alone gives LOEO $R^2 = 0.600$ (RMSE 138 mm), adding n raises it to 0.857 (RMSE 74 mm) and adding $\ln(1+I)$ to 0.882. A gradient-boosting model restricted to the same three

inputs reaches LOEO $R^2 = 0.778$, below the law, so the advantage of the compact form is not that it uses more information but that its functional form (power law in W, exponential in n) extrapolates across experiments better than piecewise-constant trees, which cannot extrapolate beyond the range of the training experiments. (This comparison uses three inputs and 15 folds and is not directly comparable with the full-library benchmark of Table 13.)

Table 12 applies Eq. (10) to three combinations of water input and irrigation in three soils. At equal input, ET is about 35 % lower in sand than in silty clay loam (263 versus 406 mm at $W = 400$ mm), because water not held in the root zone drains; the implied drainage plus storage change $W - ET$ is 137 mm in sand and negative-to-small in the fine soils, in line with the memory results. The example is illustrative; its uncertainty is the RMSE of 66–69 mm.

Table 12: ET (mm yr^{-1}) predicted by Eq. (10) for three water inputs W with irrigation I.

	$W = 400$, $I = 0$	$W = 650$, $I = 200$	$W = 1000$, $I = 300$
S03 sand ($n = 2.68$)	263	420	561
S01 coarse loam ($n = 1.353$)	391	624	834
S10 silty clay loam ($n = 1.23$)	406	648	865

Table 13 collects all sparse laws that were fitted. Transpiration is moderately predictable by a two-term law ($\ln T = -1.329 + 1.127 \ln W - 0.501\, n$; LOEO $R^2 = 0.641$, equal to gradient boosting 0.642). Canopy evaporation is predicted at LOEO 0.668 (gradient boosting 0.691) by $\ln E_c = -13.3 - 0.470 \ln W + 3.157 \ln P_r + 4.240\, I/W$, whose large coefficients reflect the narrow range of E_c and should not be interpreted mechanistically. Crop yield is predicted at LOEO 0.604 (gradient boosting 0.650). In contrast, soil evaporation is poorly predicted by sparse terms ($\ln E_s = 3.271 - 0.770\, n - 0.380$ mulch $+ 0.462 \ln P_r$; LOEO 0.308 versus 0.547 for gradient boosting; LOSO 0.348 versus 0.476), indicating that interactions that the library does not contain control E_s. The drainage fraction (DPF $= D/W$), the natural target given the PCA result, is the worst: the best sparse model with exogenous predictors has a LOEO R^2 of 0.43 and a LOSO R^2 of 0.22, against 0.50 (LOEO) for gradient boosting, and allowing partition predictors in the library raises gradient boosting to 0.59 but does not change the sparse law. The shortfall is a negative result for compact drainage laws.

Table 13: Sparse laws and benchmark. cross-validated R^2 on the log scale. Selected terms in order of selection.

Target	Selected terms	Sparse LOEO	Sparse LOSO	GBM LOEO	GBM LOSO
$\ln ET$	$\ln W$, n, $\ln(1+I)$	0.898	0.904	0.876–0.879	0.881–0.894
$\ln T$	$\ln W$, n	0.641	0.652	0.642	0.659
$\ln E_s$	n, mulch, $\ln P_r$	0.308	0.348	0.547	0.476
$\ln E_c$	$\ln W$, $\ln P_r$, I/W	0.668	0.669	0.691	0.806
ln yield	$\ln W$, $\ln(1+I)$, $\ln P_r$, mulch	0.604	0.617	0.650	0.743
ln DPF	$\ln(W/TAW)$, $\ln\alpha$, I/W, ΔT	0.433	0.217	0.498	–
$\ln R$	$\ln(1+I)$, n, cap, ΔT	0.41[†]	0.43[†]	0.43–0.47[†]	–

[†] R^2 on the log scale for R; level-scale R^2 is 0.30 (sparse) and 0.50–0.51 (gradient boosting).

Because the balance holds to within a few percent in 15 of 17 experiments, drainage can be derived by closure as $\hat{D} = W - \widehat{ET}$, which converts a hard prediction problem (drainage) into one with a good compact law (ET). On the 8 054 runs outside E8 and E9 this yields $R^2 = 0.69$ and RMSE 64 mm against Daisy drainage (Fig. 7f), and negative predicted drainage in 4.8 % of runs. The accuracy is limited by the relative size of the quantities and is not uniform (Table 14): the mean absolute error is smallest, 21–28 mm, for Daisy drainage between 10 and 200 mm (relative error 19–66 %), but the closure-derived value overpredicts by 32 mm on average for drainage below 10 mm (1 944 runs) and *underpredicts* by 69 mm and 308 mm for drainage of 200–500 and above 500 mm. The ET law is smooth in W and cannot represent the large drainage events, so closure-derived drainage is a useful consistency check for moderate drainage and not a predictor of extremes.

Table 14: Closure-derived drainage $W - \widehat{ET}$ (LOEO ET law) against Daisy drainage by drainage class (E8, E9 excluded). Bias = predicted minus Daisy.

Daisy drainage (mm yr^{-1})	Runs	Bias (mm)	Mean absolute error (mm)
< 10	1944	+31.7	39.7
10–50	2631	+8.2	21.0
50–100	1736	−2.2	25.6
100–200	1284	−13.7	28.3
200–500	377	−68.8	104.0
> 500	82	−308.1	318.6

3.5 Irrigation compensation is regime-specific

A natural candidate invariant is the compensation of rainfall changes by irrigation, κ (Eq. (7)). It is not universal (Fig. 8, Table 15). In single-soil lattices the slope of W on rain is 0.58 ± 0.02 in E11 ($\kappa = 0.42$) and 0.33 in E12 and E14 ($\kappa = 0.67$), whereas in E16 the pooled κ across the ten

soils is 0.159 (bootstrap 95 % interval 0.134–0.181) and the per-soil values range from -0.014 (sand) to 0.303 (coarse loam). Within E16, κ falls from 0.193 to 0.123 as E_p is raised from 0.3 to 1.2, from 0.188 to 0.129 with warming from $+2$ to $+6$ °C, and from 0.211 to 0.107 with trigger suction from 50 to 2000 cm, but rises from 0.079 to 0.241 with irrigation dose from 1.2 to 30 mm. Compensation therefore depends on the scheduling rule and soil, and a single value of κ should not be used in water-allocation arguments.

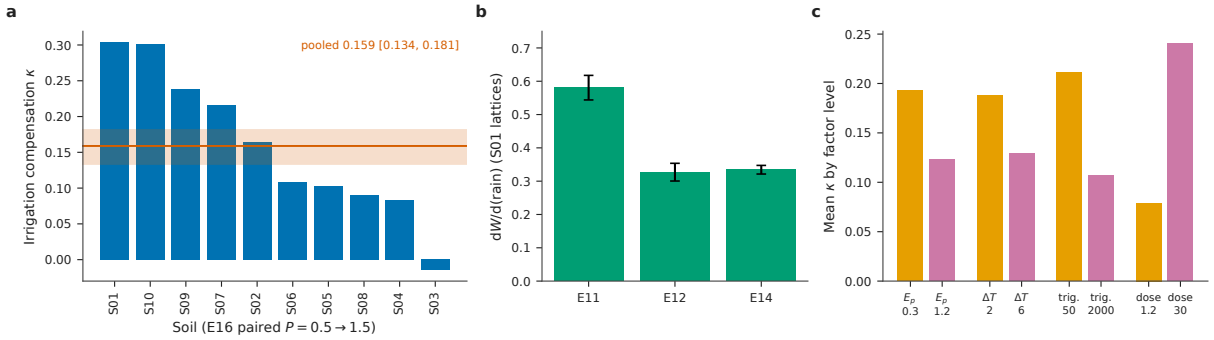


Figure 8: Irrigation compensation κ is not universal. (a) Per-soil κ in E16 (paired $P = 0.5 \rightarrow 1.5$) with the pooled estimate and bootstrap interval. (b) Slope of water input on rain in single-soil lattices (95 % intervals). (c) Mean κ by factor level in E16.

Table 15: Compensation $\kappa = -\Delta I/\Delta P_r$ and the response of ET and drainage to rain, per soil in E16.

Soil	κ	$\Delta ET/\Delta P_r$	$\Delta D/\Delta P_r$	Pairs
S01	0.303	0.512	0.186	32
S02	0.164	0.346	0.495	8
S03	-0.014	0.473	0.550	32
S04	0.084	0.473	0.453	32
S05	0.102	0.551	0.358	32
S06	0.108	0.619	0.281	32
S07	0.216	0.542	0.246	32
S08	0.090	0.613	0.308	32
S09	0.238	0.515	0.253	32
S10	0.302	0.477	0.237	32

4. Discussion

4.1 Contributions of a validated discovery protocol and a quantified accounting gap

The paper contributes a discovery protocol in which every stage is tested against a known answer, and a first application that exposes an unclosed term in a widely used process model. The pipeline is blind to variable roles; it recovers the water balance as the dominant invariant, with

a permutation null (smallest singular value 5.9 against 33.3 for permuted data), and this positive control is what licenses the interpretation of every other result. Applied to the evapotranspiration partition, the same pipeline quantifies a term that the reported components do not carry, $R \approx 164\,\mathrm{mm\,yr^{-1}}$ or about 30 % of ET, and a storage non-closure in two experiments (median 16 %). A compact three-term law then predicts ET across withheld experiments and withheld soils (LOEO $R^2 = 0.898$, LOSO 0.904) as accurately as gradient boosting (0.879 and 0.894). Quantifying a closure gap has a long record as a productive step in flux science: the energy-balance closure problem at eddy-covariance sites, in which turbulent fluxes typically fall 10–30 % short of available energy, was characterised across a network and then drove advances in measurement and in the interpretation of flux data (Foken, 2008; Wilson et al., 2002). The unattributed term R has the same function for model-output accounting: it converts an implicit assumption (that reported components are the total) into a measured quantity that can be traced to a pathway. Stating the recovered invariant for what it is, the known mass balance returned by a blind method, is the credential that makes the new findings credible in a field where discovery claims are often overstated (Nearing et al., 2021; Rudin, 2019).

Evapotranspiration partitioning is a central topic, from global studies that infer transpiration fractions (Jasechko et al., 2013; Schlesinger and Jasechko, 2014) to field studies of mulch effects (Balwinder-Singh et al., 2011). If a model's reported ET components omit a third of the flux, then a partition fraction computed as $T/(T + E_s + E_c)$ is larger than T/ET by a factor $1/(1 - 0.30) \approx 1.4$ on average, an error comparable with the differences between management treatments that are routinely reported. The size of R also helps to explain why the closure-derived and component-based estimates of soil evaporation differ. For users of Daisy and similar models, the practical recommendation is to compute ET as the model's own total and to report R alongside the components, and to document, in the model output specification, exactly which processes enter the total (Holbak et al., 2026).

4.2 A map of predictability across the flux budget

The ET law works because ET is the large, smooth, sub-linear function of supply that the water-balance literature has described for decades, and because the model's ET is energy- and storage-limited (Budyko, 1974; Milly, 1994; Zhang et al., 2001). The drainage fraction is the small residual of two large numbers, it is strongly nonlinear in the wetness of the profile, and it depends on thresholds (field capacity, trigger) that a log-linear library cannot represent (Rodriguez-Iturbe

and Porporato, 2004); gradient boosting captures some of this and still only reaches $R^2 \approx 0.5$ in LOEO. This locates the information boundary: the content that drainage needs lies beyond the annual-flux library (e.g. intra-annual timing, event intensity, antecedent state). The companion paper shows that antecedent state is indeed decisive for drainage in coarse soils.

The three-term law is competitive with gradient boosting under LOEO and LOSO, which supports the view that accuracy does not always require opacity (Rudin, 2019). The law does not extend to the regimes in which the physics it omits is first-order (initial-state experiments, dynamic hydraulic properties), and the gradient-boosting benchmark fails there too. This agreement is itself informative: it shows that the boundary is set by the information in the library and not by the model class, which tells a modeller which targets can be emulated by compact laws and which need timing and state variables.

Equation discovery in soil hydrology has so far targeted the governing partial differential equation, notably the Richards equation recovered from soil-moisture time series by group-sparse regression (Song et al., 2022, 2025), and deep-learning studies have imposed water-balance closure as a constraint on forecasts (Li et al., 2024). The present paper is complementary and differs in its object and its emphasis: it works with annual flux budgets, tests whether a pipeline recovers a known balance before interpreting any other result, audits the closure of the partition that supplies its targets, and validates by withheld regimes. The present approach shares with SINDy-style discovery (Brunton et al., 2016) the use of sparse regression over a library, but differs in two respects: it targets steady annual relations rather than dynamics, and it places the validation burden on leave-one-regime-out testing (Roberts et al., 2017). It shares with physics-informed learning (Karniadakis et al., 2021; Raissi et al., 2019) the emphasis on conservation, but imposes it as a *diagnostic* (closure residual) rather than as a constraint (Beucler et al., 2021); applying it as a constraint would have hidden R, which is the finding most relevant to model users. Related work on learning conserved quantities (Iten et al., 2020; Liu and Tegmark, 2021) would be a natural extension to daily dynamics.

Emulators are used to make process-model knowledge available in fast decision tools (Asher et al., 2015; Razavi et al., 2012). The results here indicate which emulator targets are safe: ET and transpiration are predictable from compact laws; drainage and soil evaporation are not, and should be emulated with care or by closure.

A software-level audit would resolve the identity of R directly. Four questions, in order of cost: (i) Which state variables are summed into the output called “evapotranspiration” in the log

file used here, and which into the three reported components? (ii) Is evaporation from surface ponds, mulch and litter reported in a separate column that was not exported in the ensemble? (iii) Does the canopy-evaporation column represent interception only, excluding evaporation of water retained on residue? (iv) Is the storage term in the model's own water balance consistent with the exported depth and boundary definitions (for example, is the 1 m drainage column the flux at the bottom of the root zone or the soil column)? If the answer to (ii) is yes, the unattributed term would be explained and the diagnostic here would have served its purpose as an audit.

Random k-fold cross-validation of the same models would have produced much higher scores, because runs inside an experiment are close neighbours in factor space. To illustrate the effect on the present data: a model that predicts $\ln ET$ from $\ln W$ alone has LOEO R^2 of 0.80 on the full data and 0.60 when E8 and E9 are removed. Leave-one-experiment-out removes entire designs, so the model must extrapolate to unseen combinations of factor ranges, which is exactly what a "law" is supposed to do (Roberts et al., 2017). The leave-one-soil-out scores answer a second question: would the law hold for a soil not used to fit it? That the two scores agree for ET (0.898 and 0.904) is evidence that the law is not driven by a single experiment or soil, but it is not evidence of validity in other climates. A nested evaluation, in which the forward selection was repeated inside each training set of 16 experiments and scored only on the held-out experiment, gives a pooled $R^2 = 0.895$, against 0.898 for the single-level score, and selects the same three terms in 15 of 17 outer folds (the two-term set $\ln W$, n in the other two); the optimism from selecting on the evaluation score is therefore small for this law.

The sub-linear exponent $\partial \ln ET / \partial \ln W = 0.646$ can be compared with the elasticity of Budyko-type curves. For a Budyko curve $ET/P = F(\Phi)$ with aridity index $\Phi = PET/P$, the elasticity of ET to P lies between 0 (energy limit) and 1 (water limit) and is about 0.5–0.8 at intermediate aridity (Budyko, 1974; Milly, 1994; Zhang et al., 2001). The fitted value is within that range. Unlike the Budyko approach, which treats catchment properties via a single parameter, the present law includes a soil term (n): the fitted coefficient implies that a unit increase in n reduces ET by 26 % ($1 - e^{-0.299}$) at fixed W, which can be read as the dependence of the effective Budyko parameter on soil hydraulic uniformity. The law is not claimed to be a Budyko curve; the mapping would require PET as a variable, and PET is constant within the weather series used.

A sequential protocol follows from these results and is offered as a general template for discovery studies in hydrology. The first element is a positive control: the pipeline is applied to a system whose governing balance is known, and it must recover that balance with a null-referenced

uncertainty, as the permutation test and bootstrap do here for the water balance (Efron and Tibshirani, 1993; Good, 2005). The second is a closure audit of every partition used as a target, since a decomposition that omits a term will propagate that omission into every law fitted to it, as the unattributed term R propagates into the transpiration and soil-evaporation fractions. The third is regime-wise validation, in which whole experiments, soils or climates are withheld, because random folds inside a designed ensemble measure interpolation between neighbouring factor combinations and not transfer to a new regime (Roberts et al., 2017). The fourth is a comparison with a flexible benchmark, reported with the cases in which the benchmark is better, so that the interpretability premium can be read directly (Friedman, 2001; Rudin, 2019). The fifth is a stated domain of validity for each law, expressed in the experiments and soils for which the leave-one-out score exceeds a stated threshold.

The unattributed term R also defines a concrete verification programme for the process model. Three comparisons would identify it. First, an ablation in which individual surface and canopy evaporation pathways are switched off in turn would show which pathway removes R from the balance. Second, a tracer-style accounting that records every term added to the evaporative flux in the model log, with its state variable of origin, would establish whether R is a reporting convention or a physical pathway. Third, a comparison across structurally different models (Richards-based, capacity-based and radiation-driven) on the same designed ensemble would show whether a non-closing partition is a feature of one model family or a property of how evapotranspiration is partitioned in general (Jones et al., 2003; Keating et al., 2003; Šimůnek et al., 2008; Steduto et al., 2009). The experimental data that would constrain the physical interpretation are lysimeter and eddy-covariance records in which evaporation of intercepted and ponded water is measured separately from transpiration (Good et al., 2015; Jasechko et al., 2013; Schlesinger and Jasechko, 2014).

Beyond hydrology, the same sequence of controls is relevant to other settings in which a regression of a conserved quantity on its drivers is read as a discovered relation. Sparse regression over a library is attractive because the result is an equation that can be examined, dimensionally checked and compared with theory (Brunton et al., 2016; Champion et al., 2019; Udrescu and Tegmark, 2020). Its reliability, however, depends on the design of the data. In an ensemble built by crossing factors, several drivers are correlated by construction, and the selection path of a forward algorithm reflects that correlation; the leave-one-experiment-out score then serves two roles, as selection criterion and as validation, which is why the LOSO scores, which are less

affected by selection, are reported alongside it. The practical consequence is that the coefficients of a discovered law should be reported with the validation regime in which they were obtained, and that a law should be described as discovered only for the regimes in which it is validated.

Finally, the results bear on the design of the emulators and decision tools that draw on process-model ensembles. A tool that reports evapotranspiration components should state whether they sum to the total; one that reports drainage should state whether it is simulated or derived by closure, and the evidence here shows that closure-derived drainage is accurate in sign and rank order but under-estimates large drainage. Emulators built on such ensembles should also be tested against held-out simulations (Asher et al., 2015; Razavi et al., 2012; Wilkinson et al., 2016).

4.3 Domain of validity of the discovered laws

Each law is reported with the regimes in which it was validated, and each boundary defines the next test.

1. *Process-model domain.* The results describe Daisy output. R is a property of the model's accounting; its counterpart in nature is measured in lysimeters and eddy-covariance records that separate intercepted and ponded water from transpiration.
2. *Failed runs.* The 149 excluded runs (87 % in S02) form a pattern that identifies them as solver failures (Section 2.2); re-running with tighter tolerances is the confirming step, and S02 is under-represented at large doses and extreme rain scales.
3. *Model and weather.* One model and one climate; the laws are conditional on both, and the leave-one-regime-out protocol is designed to be repeated on further models and climates.
4. *Selection.* Forward selection used the LOEO score; the nested evaluation, with selection repeated inside each training set, gives $R^2 = 0.895$ against 0.898 for the ET law, so the selection optimism is small. LOSO values are less affected by selection.
5. *Dependence of runs.* Coefficient standard errors assume independent runs; cross-fold variation is the reported guide to uncertainty.
6. *Soil covariates.* Ten soils; n carries several correlated soil properties.
7. *Identification of R.* Its surface-layer origin is inferred from its driver dependence, and the four-question audit of the model code in this section settles it.
8. *Aggregation.* Twenty-year means average over seasonality and event structure; the companion work addresses antecedent state.
9. *Benchmark reproducibility.* The gradient-boosting benchmark is run with a fixed random

seed. Across unseeded repeats the LOSO score varied by up to 0.013, so differences below about 0.02 between a law and the benchmark are not interpreted.

(1) Begin with a positive control with a known answer. (2) Use a permutation null for any decomposition-based invariant. (3) Validate across *regimes* (experiments, soils, climates), not only across random folds. (4) Audit partitions for closure before using them as targets. (5) Compare with a flexible benchmark and report where the benchmark wins. (6) State a domain of validity for each law. (7) Publish failures.

5. Conclusions

A blind linear-invariant search recovers the water balance from a flux matrix with every coefficient within 5 % of the exact value, calibrated against a permutation null, and flags two accounting gaps: an unattributed evapotranspiration term averaging 164 mm yr^{-1} (30 % of ET) that depends on irrigation method, evaporation factor and mulch, and a storage non-closure of 16 % in the two initial-moisture experiments. Compositional PCA shows that a single drainage–evaporation axis dominates water partitioning, with a variance share (52–89 %) that depends strongly on how zero drainage is handled. A three-term law, $\ln ET = 2.503 + 0.646 \ln W - 0.299\, n + 0.029 \ln(1 + I)$, matches gradient boosting in leave-one-experiment-out ($R^2 = 0.898$) and leave-one-soil-out ($R^2 = 0.904$) validation, while sparse laws for drainage fraction, soil evaporation and the unattributed term do not, and irrigation compensation is not a universal constant. The conclusions are limited to a single process model and weather series, and the discovered "law" is the known balance together with a diagnosed accounting gap.

Data and code availability

Data. The complete Daisy ensemble outputs (all 10 190 runs, raw and derived) are archived on Zenodo, doi: 10.5281/zenodo.23043626. The raw run table used here, and a linked list of every data file used, are in the Data folder of the project repository (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/tree/main/Data).

Code. The analysis and figure scripts are in the Code folder (https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Preci

sion-Irrigation-Mulching-Feedback-Loops/tree/main/Manuscripts). The folder Manuscripts/reproducibility is a self-contained package: pinned software versions (Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5, scikit-learn 1.9.1, statsmodels 0.15.0, Matplotlib 3.11.2, Pillow 12.3.0), fixed random seeds in every stochastic step, SHA-256 checksums, and one command (run_all.sh) that regenerates all results and figures from the ensemble and then compares every regenerated value with the archived result and with the headline numbers printed in the manuscript. Each table and figure names the script that produced it (Appendix B). Daisy itself was not re-run for this package, no field measurements are used, and small intermediate arrays used only by figures are archived rather than regenerated; these limits are listed in README_REPRODUCE.md.

Author contributions

N.H.: conceptualisation, analysis, software, visualisation, writing.

Funding

No funding was received for this work.

Competing interests

The author declares no competing interests.

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A. Supplementary information

The analysis was exploratory and was not pre-registered. The main analytical decisions were: the four controls (permutation null, bootstrap, LOSO/LOEO, gradient-boosting benchmark); the forward-selection stopping threshold (0.003); log-targets; the exclusion of E8 and E9 as a sensitivity analysis for the invariant search and for closure-derived drainage; and the treatment of R as a remainder. The candidate library was assembled by the author with knowledge of the ensemble and of the companion hydraulic-memory results (for example, the choice to include n, $\ln \alpha$ and $\ln W/TAW$), which increases researcher degrees of freedom; readers should treat LOEO and LOSO scores as indicative rather than as confirmatory error rates.

The seven-flux search also returns weights on transpiration, soil evaporation and canopy evaporation in its smallest direction (rain 0.896, irrigation 0.834, ET -1, transpiration 0.367, soil evaporation 0.121, canopy evaporation -0.122, drainage -0.933), which is not a clean invariant: because $T + E_s + E_c + R = ET$ and R is not a column, the seven-column matrix has an additional near-dependence that mixes components. This is why the four-flux system is the clean positive control.

Table 16: Columns of the ensemble used in this paper.

Column	Unit	Definition
`experiment`, `treatment`, `run_id`	–	experiment code, treatment string encoding factor levels, unique run identifier
`soil`	–	S01–S10
`rain_mm`, `irrigation_mm`	mm yr^{-1}	20-year mean annual rainfall and irrigation
`ET_mm`	mm yr^{-1}	model evapotranspiration (total)
`transpiration_mm`, `soil_evap_mm`, `canopy_evap_mm`	mm yr^{-1}	reported components of ET
`drainage_1m_mm`	mm yr^{-1}	flux across the 1 m boundary (negative = net rise)
`water_stress_d`	d yr^{-1}	days of water stress
`yield_Mg_ha`	Mg ha^{-1}	grain/biomass yield
`N_fertiliser_kg_ha`, `N_leached_kg_ha`	kg ha^{-1}	nitrogen applied and leached
`IE_kg_m3`, `DPF`, `NLF`	–	irrigation efficiency, drainage fraction D/W, N-leaching fraction (definitional; excluded as targets)
`P`, `dT`, `Ep`, `trig`, `z`, `amt_mm`, `vff`, `cap`, `Nrate`	various	factor levels (rain scale, warming, evaporation factor, trigger, sensor depth, dose, bare fraction, mulch capacity, N rate)

DPF$= D/W$ is exact by construction and its "prediction" is therefore the same task as predicting D; NLF and IE are ratios of other columns. They are not used as independent evidence in any result.

B. Reproducibility map

Table 17: Result-to-script map.

Result	Script / output
Ensemble table and run counts	`make_data.py` → `data/daisy_ensemble.csv.gz`
SVD, permutation null, closure (Eq. (8), Fig. 3)	`p2_a.py`, `p2_laws.py` → `p2_a.json`, `p2_laws.json`
Unattributed term R (Eq. (9), Fig. 4)	`p2_R.py` → `p2_R.json`
clr-PCA (Fig. 5, Table 7)	`p2_a.py`, `p2_figs.py`, `p2_fig4.py` → `p2_pca.json`, `p2_pca_sens.json`
Sparse laws (Table 13, Eq. (10))	`p2_et.py`, `p2_dpf.py`, `p2_figs2.py`, `p2_nested.py` → `p2_et.json`, `p2_dpf.json`, `p2_etlaw.json`, `p2_nested.json`; exclusion sensitivity `p2_s02_excl.py` → `p2_s02_excl.json`
Cluster bootstrap, law without E8 and E9, descriptive statistics (Table 2) and R by experiment (Table 6)	`p2_robust.py` → `p2_robust.json`
Compensation (Fig. 8)	`p2_laws2.py` → `p2_laws2.json`
Figures	`p2_figs.py`, `p2_figs2.py`, `p2_fig4.py`, `p2_fig6.py`, `p2_fig7.py`, `p2_figfail.py`, `p2_fig1_new.py`
All steps, checks and limits	`run_all.sh`, `compare_results.py`, `audit_numbers.py`, `README_REPRODUCE.md`